

INFOSYS | SPRINGBOARD

Virtual Internship 7.0

COMPLETION REPORT

AI-Based Industrial Maintenance and Process Optimization For Smart Manufacturing

Team Details

Project Name: AI-Based Industrial Maintenance and Process Optimization For Smart Manufacturing

Start date: 26 - 06 - 2026

Names:

1. Mahesh
2. Nagaroopini D
3. Natarani Nirupa
4. Neelima Kesanak
5. Priyanka H C
6. Thamma Pavani

Internship Duration: 45 Days

Executive Summary

AI Based Industrial Maintenance and Process Optimization is an AI-driven dashboard system built during the Infosys Springboard Virtual Internship 7.0. The project addresses a common industrial problem reactive maintenance, where machines are repaired only after they fail, causing unplanned downtime and elevated costs.

Using the AI4I 2020 industrial sensor dataset, the team built a complete pipeline: cleaning and scaling real sensor readings, training and comparing three machine-learning classifiers (Logistic Regression, Random Forest, and XGBoost), and deploying the best-performing model behind a live, interactive Streamlit dashboard. The dashboard renders a simulated 10-machine fleet as color-coded risk cards, automatically alerts when any machine's predicted failure risk crosses a critical threshold, and includes a Groq-powered AI Assistant that answers natural-language questions grounded in the live fleet state.

The team also implemented a secured employee login and self-registration system — with bcrypt-hashed passwords, cookie-based sessions, and sign-up restricted to pre-approved employee emails — so the dashboard is accessible only to verified users, not the public internet.

Beyond the technical build, the team identified and transparently addressed a data-leakage issue in the source dataset, worked through several real-world environment and deployment challenges, and documented the full journey in this report. The result is a practical, honestly-evaluated predictive-maintenance system with a clear path for further enhancement.

1. Project Title

AI Based Industrial Maintenance and Process Optimization For Smart Manufacturing

2. Project Objective

AI Based Industrial Maintenance and Process Optimization For Smart Manufacturing is an AI-driven system designed to predict machine failures from live sensor readings, before a breakdown occurs. Its core objectives are to:

- Predict machine failure risk using machine-learning models trained on real industrial sensor data.
- Monitor a live fleet of machines continuously through an interactive dashboard.
- Surface failure risk clearly, with automatic alerts once risk crosses a defined threshold.
- Support plain-language questions about machine health via an integrated AI assistant.
- Restrict dashboard access to verified employees only, through a secured login and sign-up system.

The project replaces reactive, after-the-fact maintenance with a proactive, data-driven approach — reducing unplanned downtime, maintenance cost, and production loss.

3. Project Description in Detail

AI Based Industrial Maintenance and Process Optimization is an AI-powered web dashboard built on the AI4I 2020 industrial dataset. The system:

1. Loads and cleans 10,000 real sensor readings — dropping non-predictive ID columns and encoding machine Type (Low/Medium/High) numerically.
2. Scales all numeric sensor features (air temperature, process temperature, rotational speed, torque, tool wear) using a fitted StandardScaler, so no single sensor dominates due to its units.
3. Trains and compares three classification models — Logistic Regression, Random Forest, and XGBoost — automatically selecting whichever scores the highest test accuracy.
4. Saves the winning model and scaler with joblib, so the dashboard loads a trained model instantly rather than retraining on every run.
5. Simulates a live 10-machine fleet on a Streamlit dashboard, rendering each machine as a color-coded card (Healthy / Warning / Critical) based on its predicted failure risk.
6. Fires an automatic alert banner whenever any machine's predicted risk crosses 70%.
7. Provides an AI Assistant, powered by Groq's Llama 3.3 70B model, that answers natural-language questions grounded in the live fleet's current state.
8. Includes a Process Intelligence tab where users can upload their own CSV data for further analysis.

9. Gates the entire dashboard behind a secured login and self-registration system — passwords are bcrypt-hashed, sessions are cookie-based, and sign-up is restricted to a pre-approved list of employee emails.

This project combines applied machine learning, data engineering, and full-stack dashboard development into a single, presentable predictive-maintenance system.

4. System Architecture & Technology Stack

The system follows a five-stage pipeline, taking a raw sensor reading all the way to a dashboard alert:

1. A sensor reading is provided (drawn from the AI4I 2020 dataset in this implementation, simulating a live IoT sensor stream).
2. The reading is preprocessed — machine Type is encoded and every numeric feature is scaled using the fitted StandardScaler.
3. The trained ML model scores the reading and produces a predicted failure-risk percentage.
4. The result renders live on the Streamlit dashboard as a color-coded card.
5. The AI Assistant and alert banner surface the result to the user in plain language.

Technology stack:

- Python — core language for data processing and model training.
- pandas & scikit-learn — data cleaning, feature scaling, and model training/evaluation.
- XGBoost — gradient-boosted classifier included in the model comparison.
- Streamlit & Plotly — the interactive dashboard framework and its charts.
- Groq API (Llama 3.3 70B) — powers the natural-language AI Assistant.
- streamlit-authenticator & bcrypt — secured login, session cookies, and password hashing.
- joblib — persists the trained model and scaler for fast dashboard startup.

5. Security & Access Control

Because the dashboard surfaces operational data about a company's machines, access is gated behind a real authentication system rather than left open:

- Log In — username and password are checked against stored credentials. Passwords are never stored as plain text; they are hashed with bcrypt before being saved.
- Session Handling — an authenticated session is kept via a secure browser cookie that expires automatically after a set number of days.

- Sign Up — only email addresses on a pre-approved employee list can self-register for an account; any other email is rejected automatically.
- UI Consistency — the login and sign-up screens use the same dark, branded theme as the rest of the dashboard, so security did not come at the cost of a polished user experience.

This design means the dashboard can reasonably be described as restricted to verified employees, not merely gated by a cosmetic screen.

6. Timeline Overview

(Based on an 8-week development plan — adjust the week numbers/dates below if your actual internship timeline differs.)

Week	Activities Planned	Activities Completed
Week 1–2	Data collection & exploratory analysis	Sourced the AI4I 2020 dataset; ran EDA (missing values, failure rate, sensor distributions).
Week 3–5	Model training & testing	Preprocessed data (dropped ID columns, encoded Type, scaled features); trained and compared Logistic Regression, Random Forest, and XGBoost.
Week 6–7	Dashboard & AI Assistant development	Built the Streamlit dashboard (Digital Twin, ML Performance, Process Intelligence, About); integrated the Groq-powered AI Assistant; added secured login/sign-up.
Week 8	Testing, documentation & final presentation	End-to-end testing, resolved environment/dependency issues, documented the project, prepared the final presentation.

6a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Defined project objectives, scope, responsibilities, and success criteria.	26-06-2026
Prototype / First Draft	Working prototype covering data preprocessing, model training, and initial dashboard.	10-07-2026
Mid-Term Review	Reviewed progress; refined model comparison approach and dashboard feature priorities.	24-07-2026

Milestone	Description	Date Achieved
Security Feature Added	Implemented secured employee login and self-registration on the dashboard.	01-08-2026
Final Submission	Completed and submitted the fully functional dashboard with reporting and documentation.	10-08-2026
Final Presentation	Delivered a demonstration covering system architecture, functionality, results, and impact.	14-08-2026

6b. Project Execution Details

The execution of the project followed a structured, iterative development approach.

The project began with requirement analysis — defining the problem (predicting machine failure before it happens), selecting the AI4I 2020 dataset, and mapping which sensor features were realistically available for prediction.

Next, data preprocessing was implemented: dropping identifier columns, encoding machine Type, and scaling every numeric feature with a StandardScaler fitted on the training split.

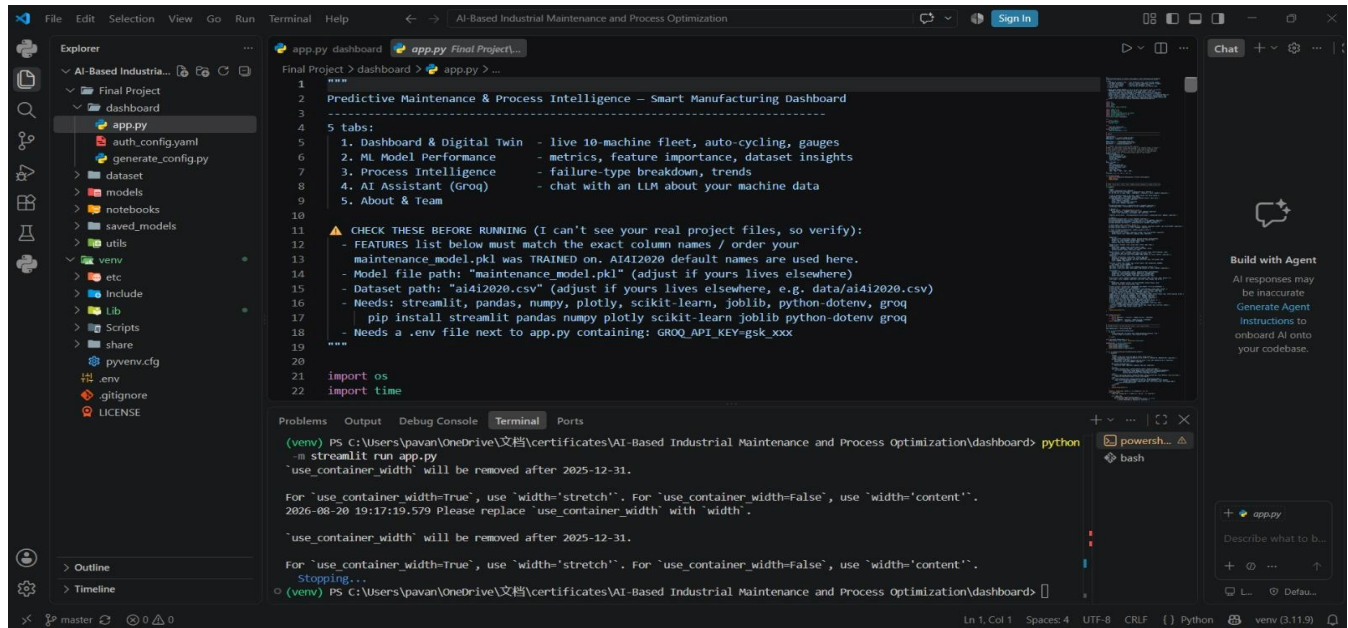
A model comparison pipeline was then built, training Logistic Regression, Random Forest, and XGBoost on the same preprocessed data and automatically retaining whichever model scored the highest test accuracy. During this stage, the team identified and addressed a data-leakage risk: the dataset's failure-mode flag columns (TWF, HDF, PWF, OSF, RNF) are only known after a failure has occurred, so live predictions deliberately feed these flags as zero rather than relying on information that wouldn't exist for a healthy, in-service machine.

In parallel, a Streamlit dashboard was developed, rendering a live 10-machine fleet as color-coded risk cards, alongside tabs for model performance metrics, process-intelligence charts with CSV upload, and an AI Assistant grounded in the live fleet snapshot via the Groq API.

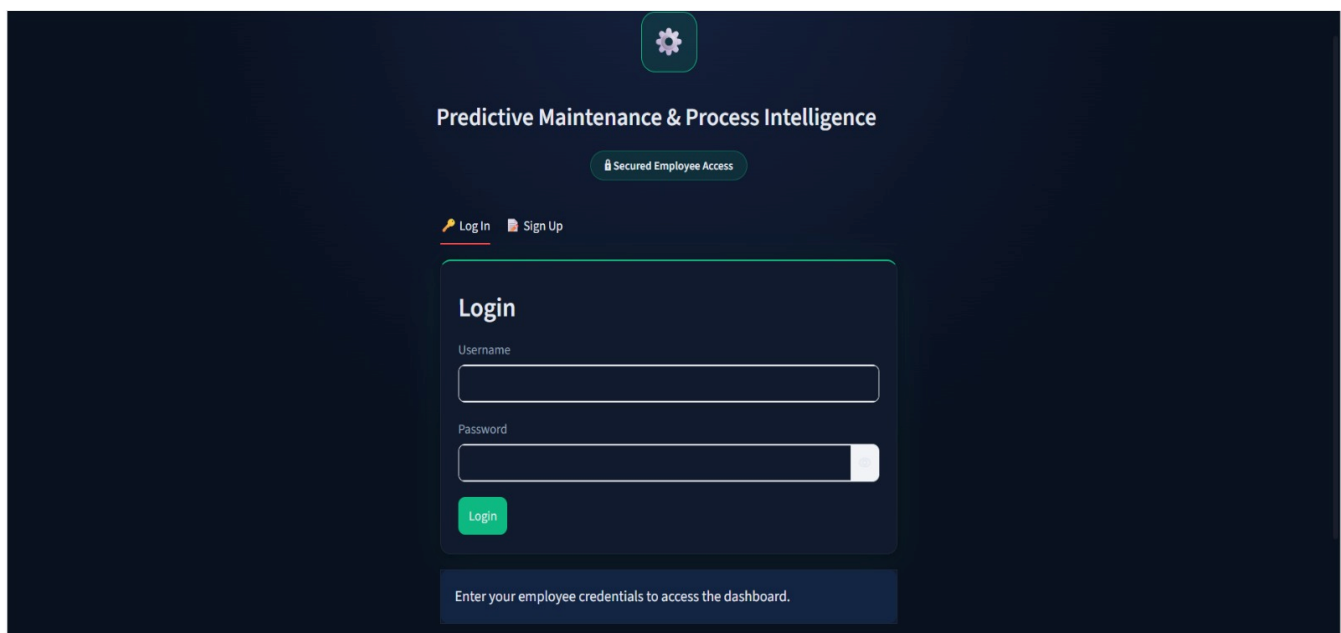
Finally, a secured login and self-registration system was added using bcrypt password hashing, cookie-based sessions, and an employee-email allow-list for sign-up, so the dashboard is accessible only to verified users. The system was tested end-to-end, several environment and dependency issues were resolved, and the project was documented and prepared for final presentation.

7. Snapshots / Screenshots

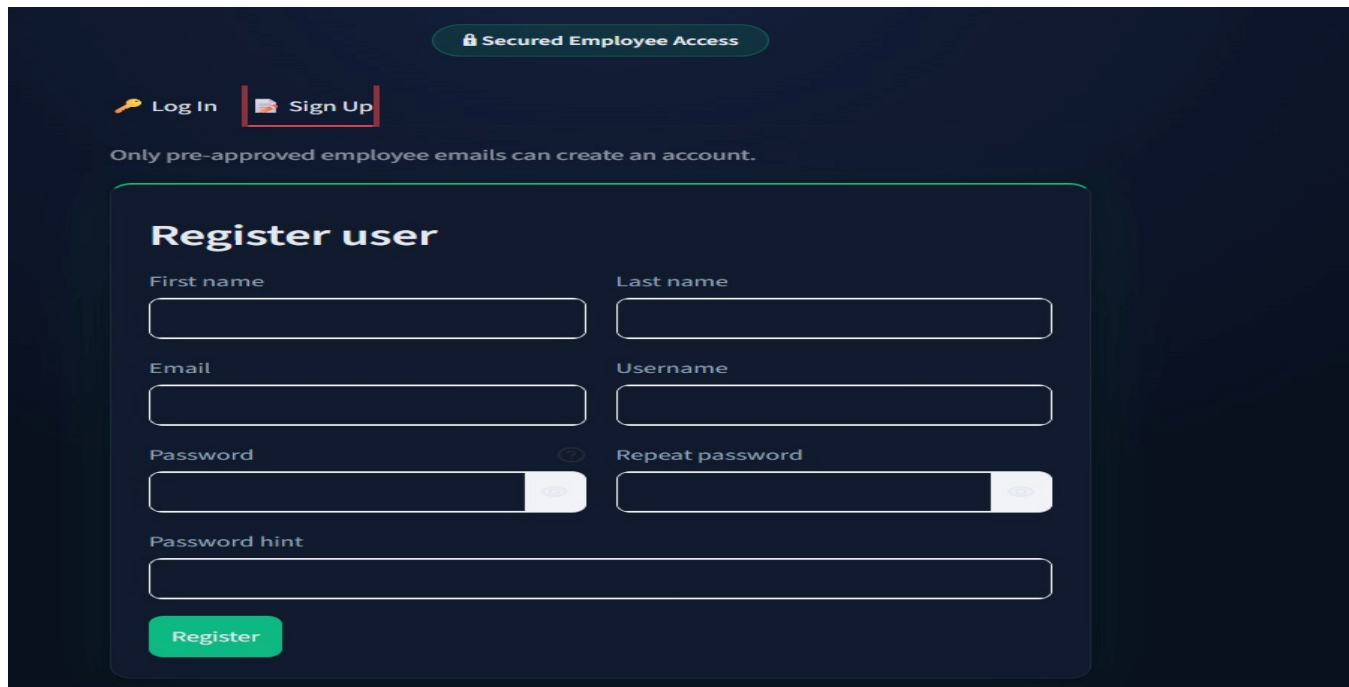
The project contains a Streamlit-based Smart Manufacturing Dashboard with modules for real-time machine monitoring, ML model performance, process intelligence, AI-based assistance, and project/team information.



The project includes a secure employee login and registration interface that authenticates users and provides controlled access to the Predictive Maintenance & Process Intelligence dashboard.



The project includes a secure employee registration module that allows pre-approved employees to create accounts by providing personal details, email, username, password, and password confirmation.



The image shows a 'Register user' form within a 'Secured Employee Access' interface. The form is titled 'Register user' and includes fields for 'First name', 'Last name', 'Email', 'Username', 'Password', 'Repeat password', and 'Password hint'. A 'Register' button is at the bottom. Above the form, there are 'Log In' and 'Sign Up' links, with 'Sign Up' highlighted. A note states: 'Only pre-approved employee emails can create an account.'

Secured Employee Access

Log In Sign Up

Only pre-approved employee emails can create an account.

Register user

First name Last name

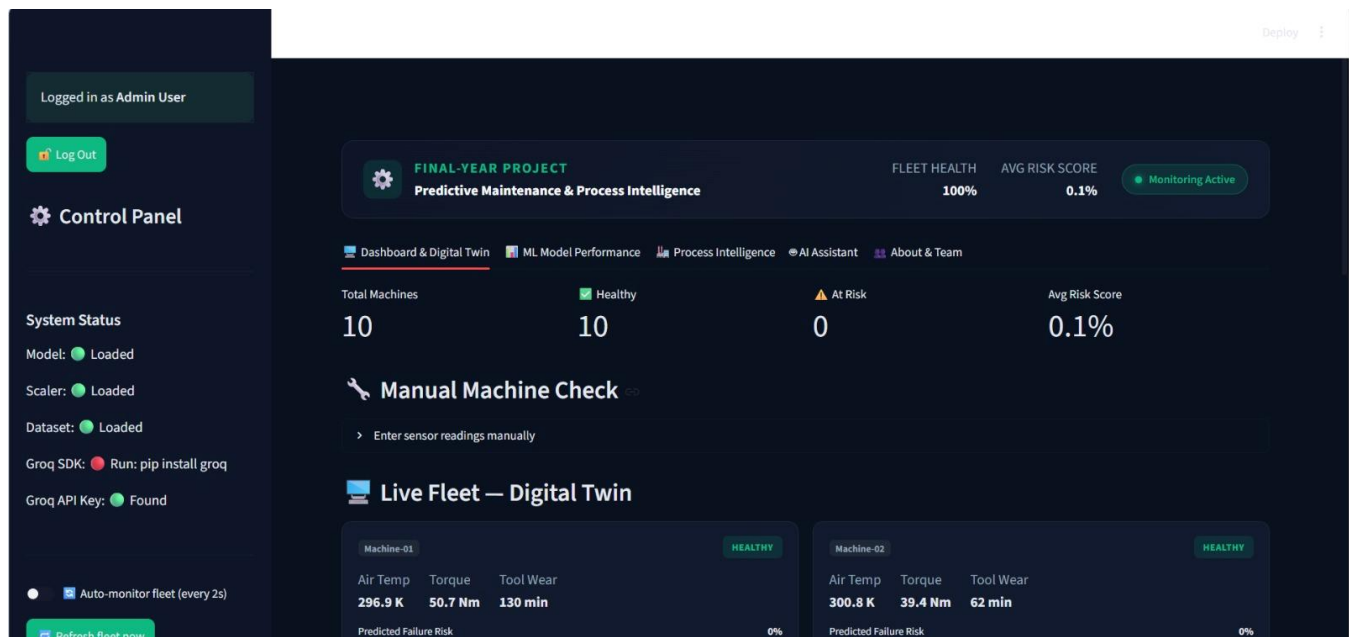
Email Username

Password Repeat password

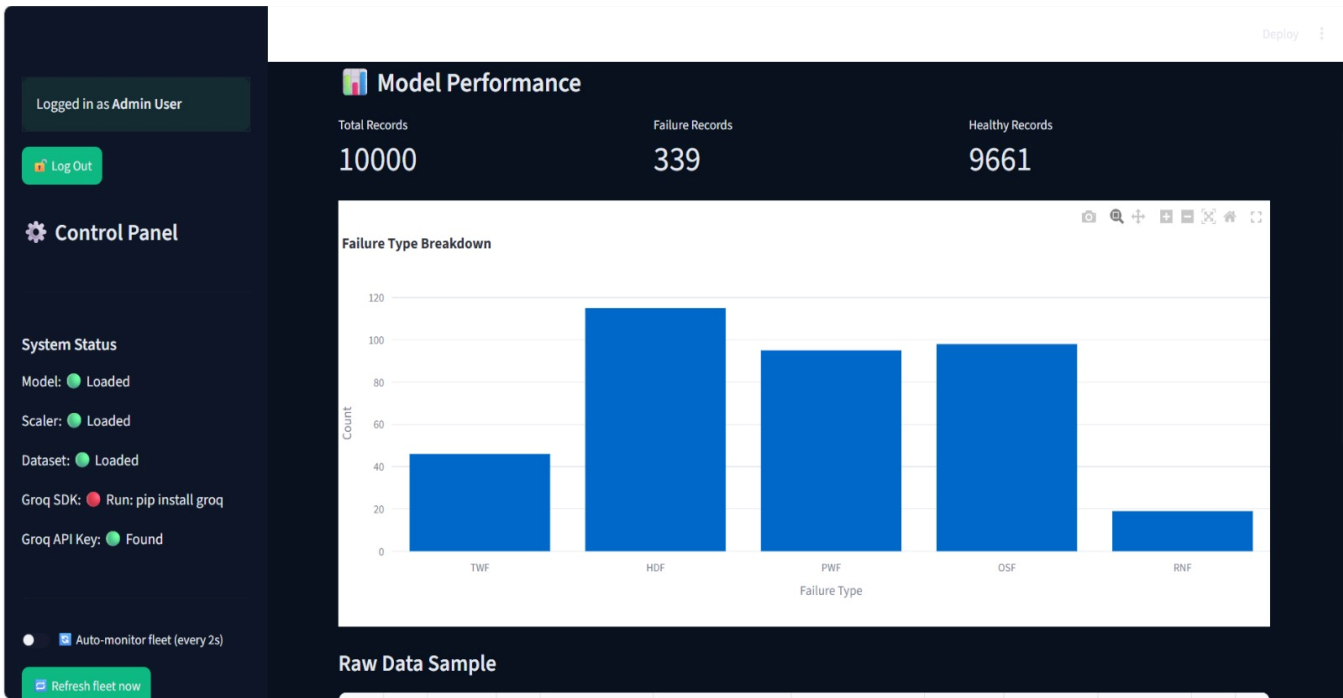
Password hint

Register

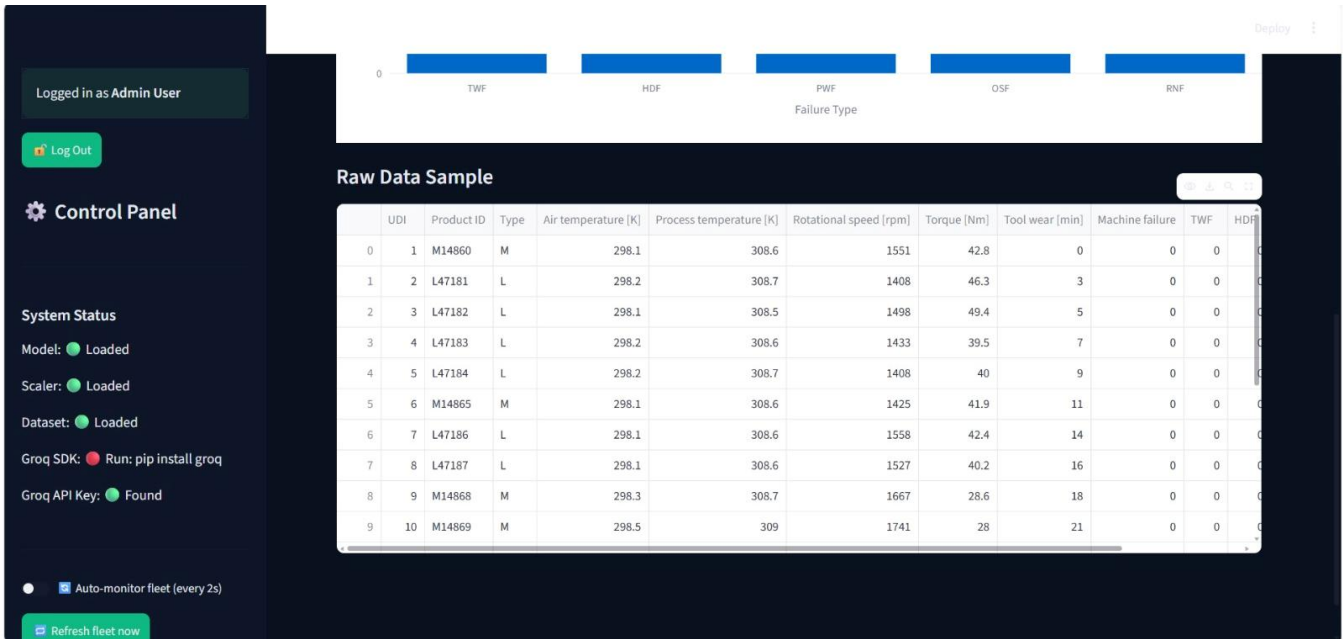
The project dashboard provides real-time monitoring of the industrial machine fleet, displaying machine health, risk scores, sensor readings, system status, and digital-twin information for predictive maintenance



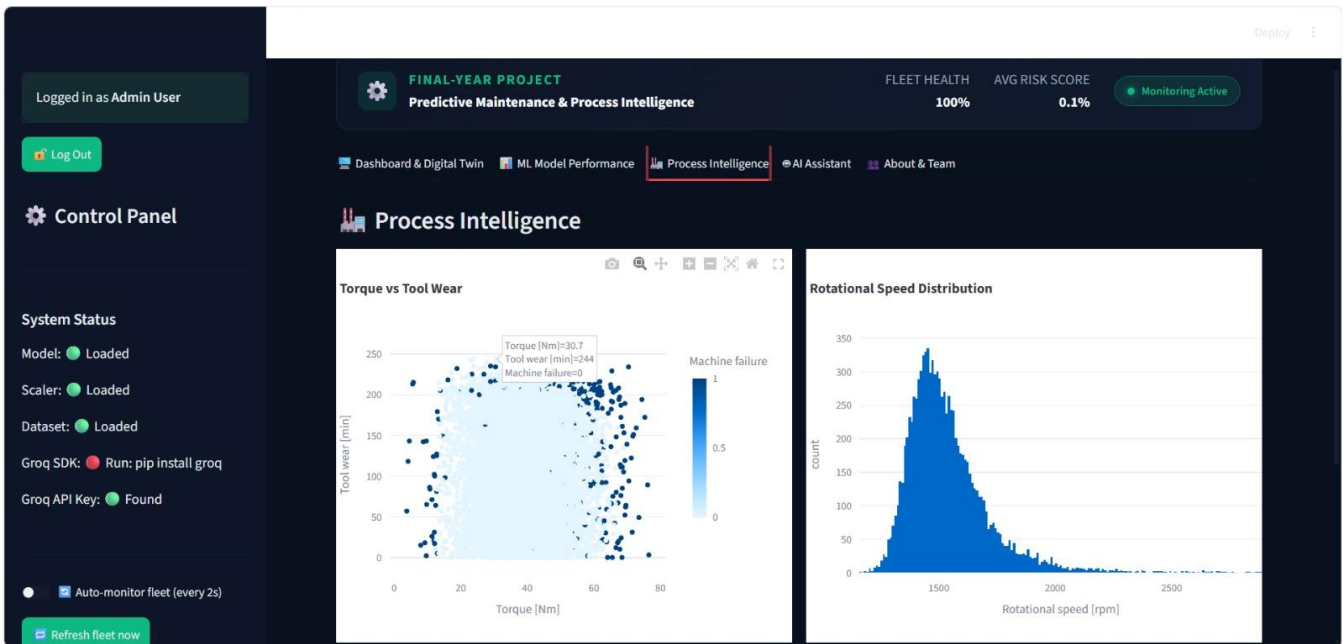
The project includes a Model Performance module that analyzes 10,000 machine records, identifies failure and healthy cases, and visualizes the distribution of different failure types for evaluating predictive maintenance performance.



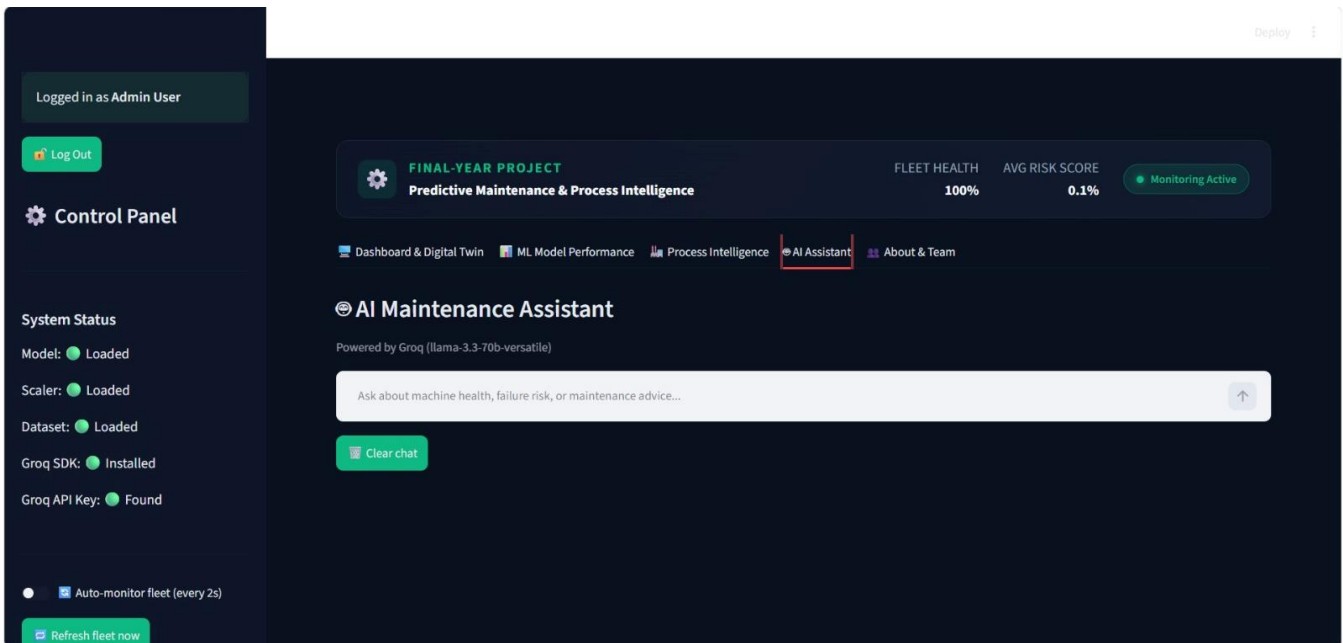
The project includes a Raw Data Analysis module that displays machine sensor data such as temperature, rotational speed, torque, tool wear, and different machine-failure indicators for detailed analysis.



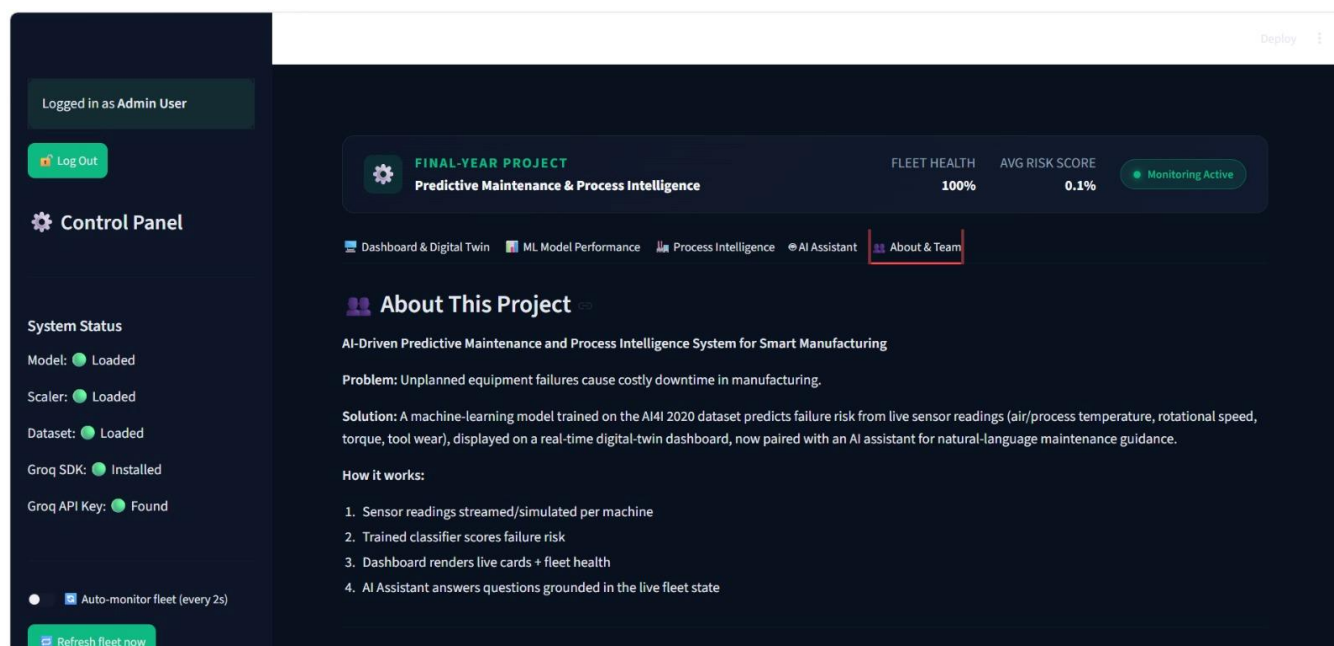
The project includes a Process Intelligence module that analyzes relationships between torque, tool wear, rotational speed, and machine failures using interactive visualizations to identify operational patterns and potential failure conditions.



The project includes an AI Maintenance Assistant powered by Groq that allows users to ask questions about machine health, failure risks, and maintenance recommendations through an interactive chatbot.



The project includes an About & Team module that explains the project's objective, problem statement, proposed AI-based solution, and the overall workflow of using sensor data, machine-learning predictions, real-time monitoring, and AI-assisted maintenance guidance.



8. Results & Impact

Dataset characteristics:

- 10,000 real sensor readings across 14 original columns.
- Failure rate of approximately 3.4% (339 failure events) — a realistically imbalanced industrial dataset.
- No missing values, which simplified preprocessing while still requiring careful feature scaling.

Model comparison approach:

Three classifiers were trained on an 80/20 train-test split: Logistic Regression, Random Forest, and XGBoost. The training pipeline automatically retains whichever model scores the highest test accuracy, removing manual bias from model selection. [Insert your team's actual printed accuracy figures here for each model.]

Anticipated operational impact:

- Earlier detection of failure risk, shifting maintenance from reactive to proactive.
- Reduced unplanned downtime and associated production loss.
- Lower emergency-repair costs compared to reactive-only maintenance.
- Faster, more confident decisions for maintenance teams via the AI Assistant.

- Controlled, auditable access to operational data through the secured login system.

9. Challenges Faced

During development, the team encountered and resolved several technical challenges:

1. **Feature-Order & Scaling Consistency** The saved StandardScaler had to be applied with the exact same feature order at inference time as during training — a mismatch here would silently produce wrong predictions without raising any error.
2. **Data Leakage in the Source Dataset** The dataset's failure-mode flag columns are only known after a failure occurs. Training on them inflated accuracy without adding real predictive value for a live, healthy machine — the team resolved this by feeding those flags as zero during live prediction.
3. **Environment & Dependency Conflicts** A newly released Starlette version broke Streamlit's internal server on startup (a GZipMiddleware incompatibility). This was resolved by pinning Starlette to a compatible version.
4. **Virtual Environment Path Breakage After Renaming** After renaming the project folder, the Python virtual environment's console-script launchers (which embed an absolute path at creation time) broke. The team worked around this by invoking Streamlit as a Python module (`python -m streamlit`) rather than through the broken launcher.
5. **Cloud-Sync Performance Delays** Working inside a OneDrive-synced folder introduced noticeable slowdowns immediately after file/folder renames, as the sync client re-indexed thousands of virtual-environment files.
6. **Balancing Security With Usability** Adding a real login/sign-up system required careful UI work so that dark theming, readable input text, and consistent styling carried through to the authentication screens, not just the main dashboard.

Despite these challenges, iterative debugging and testing allowed the system to reach stable, working performance.

10. Learnings & Skills Acquired

The development of this project resulted in significant technical and professional learning outcomes:

1. **Practical Machine Learning Evaluation** Compared multiple classifiers (Logistic Regression, Random Forest, XGBoost) and learned to recognize and correct for data leakage in a real dataset.
2. **Data Preprocessing Discipline** Learned to design a full data pipeline — cleaning, encoding, and scaling — kept consistent between training and live inference.
3. **Dashboard & UI Development** Built a multi-tab interactive dashboard in Streamlit, including live data visualization, custom styling, and session state.

4. **Secure Authentication** Implemented password hashing, cookie-based sessions, and access control restricted to pre-approved users.
5. **AI/LLM Integration** Integrated a hosted LLM (Groq's Llama 3.3 70B) to answer questions grounded in live application state.
6. **Debugging & Environment Management** Diagnosed and resolved real-world environment issues — dependency version conflicts, broken virtual-environment paths, and cloud-sync interference.
7. **Project Structure & Reproducibility** Practiced structuring a project (dashboard/, models/, utils/, saved_models/) for clarity and reuse across scripts.

Overall, the project strengthened both AI/ML technical competencies and full-stack development skills, while reinforcing the importance of honest, leakage-free model evaluation.

11. Testimonials from Team

Mahesh:

“Working on this project helped me understand how artificial intelligence and machine learning can be applied to real-world industrial problems. I contributed to the development and understanding of the predictive maintenance system and gained valuable experience in teamwork, data analysis, and project implementation.”

Nagaroopini D:

“This project gave me practical knowledge about predictive maintenance and smart manufacturing. I learned how machine sensor data can be analyzed to identify possible failures and how different modules can be integrated into a single intelligent system. It also improved my technical and teamwork skills.”

Natakarani Nirupa:

“Through this project, I gained a better understanding of machine-learning-based failure prediction and process intelligence. I learned how data visualization and analysis can help monitor machine performance and support better maintenance decisions. Working as a team also improved my communication and problem-solving abilities.”

Neelima Kesanak:

“Being part of this project helped me understand the complete workflow of an AI-driven predictive maintenance system. I learned about machine health monitoring, failure-risk analysis, and the importance of real-time data in smart manufacturing. This experience strengthened both my technical knowledge and my confidence in project development.”

Priyanka H C:

“Working on this project gave me practical experience in developing an AI-based predictive maintenance and process intelligence system. I learned how to work with industrial datasets, machine-learning models,

data visualization, and real-time dashboard development. The project also helped me improve my problem-solving, collaboration, and presentation skills.”

Thamma Pavani:

“This project provided me with valuable hands-on experience in applying artificial intelligence to industrial maintenance. I learned how sensor parameters such as temperature, rotational speed, torque, and tool wear can be used to predict machine failures. The project also taught me the importance of teamwork, coordination, and practical implementation.”

12. Future Scope

- Retrain the model without the leakage-prone failure-flag columns, for a fully honest live-prediction model.
- Connect to real IoT sensor streams instead of dataset-simulated readings.
- Add historical trend tracking per machine, so risk changes over time are visible, not just a single snapshot.
- Deploy to the cloud for remote fleet monitoring across multiple sites.
- Expand the AI Assistant to proactively suggest maintenance schedules, not just answer questions.

13. Conclusion

AI Based Industrial Maintenance and Process Optimization successfully demonstrates how machine learning and AI can shift industrial maintenance from reactive to predictive. The system analyzes live sensor readings, scores failure risk in real time, and surfaces that risk clearly to operators through an interactive, secured dashboard.

Throughout development, the team strengthened its expertise in end-to-end pipeline design — from data preprocessing and model comparison to dashboard development, AI-assistant integration, and secure access control. The project also demonstrated the team's ability to identify and honestly correct a data-leakage issue rather than presenting inflated results.

Overall, the project stands as a practical, well-documented AI solution with clear potential for further enhancement — including retraining without leakage-prone features, connecting to real IoT sensor streams, and expanding historical trend analysis.

14. Acknowledgements

We express our sincere gratitude to **Infosys Springboard** for providing an excellent learning environment, industry-level curriculum support, and access to resources that helped shape this project effectively.

We extend heartfelt thanks to **Ms. Sangeetha Mahaligam**, our mentor, for continuous guidance, constructive feedback, and encouragement throughout the development lifecycle.

We also appreciate the support from our faculty members, peers, and family for their motivation, collaboration, and valuable suggestions, which helped us deliver this project successfully.